

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

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Software

- [Review](#) 
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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configuration ([Kiss et al., 2020](#)). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site ([Figure 1](#)) and backed by a citable archived release ([COSIMA contributors, 2026](#)).

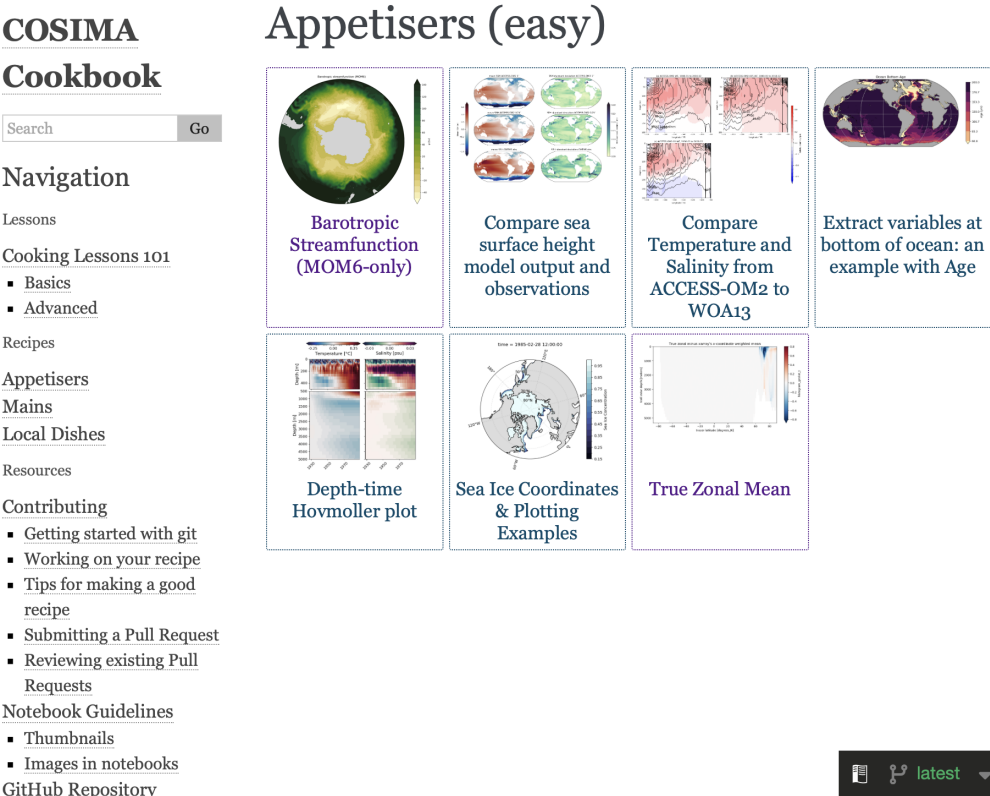


Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

32 The website sections are deliberately named using a gastronomy theme, consistent with
33 the “Cookbook” concept □□□ □□□. “Cooking Lessons 101” refers to tutorials that teach
34 generic, transferable techniques; “Appetisers” are easier recipes that provide a good entry
35 point after the tutorials; “Mains” are more elaborate and advanced analysis examples;
36 and “Local Dishes” collect recipes for regional model configurations (Barnes et al., 2024).
37 Together these sections present the documentation as a browsable collection of lessons
38 and recipes gathered into a single cookbook □.

39 Pedagogy and instructional structure are central to the project. “Cooking Lessons 101”
40 introduces generic skills such as loading model output, working with labelled arrays,
41 plotting, and interacting with shared data catalogues. These tutorials lead into domain-
42 focused “recipes”: self-contained notebooks showing how to perform concrete diagnostics
43 and analyses on ocean and sea-ice datasets. The current repository contains dozens of
44 notebooks grouped into introductory, advanced, and region-specific collections, giving
45 learners an incremental path from first contact with the data ecosystem to adaptation of
46 full workflows for their own science questions.

47 The Cookbook grew from a practical need inside the COSIMA community: the tools used
48 to analyse modern ocean model output are powerful, but the gap between package-level
49 documentation and a reproducible end-to-end workflow remains large. Users need to
50 understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate
51 high-dimensional model output, shared high-performance computing environments, and
52 domain-specific analysis conventions. The Cookbook addresses that gap by packaging
53 reusable workflows in the same medium in which users actually work.

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Author order

The first two authors contributed equally; authors from the third position onward are listed alphabetically by surname.

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